

UAV-X Resilient BVLOS Swarm

Results of all 60 simulations and the complete mathematical model

Stage 1 proof of concept v3.1 · Webots R2025a + identical headless core · PUSHPAK Grand Challenge 2026

Simulations	Passed every constraint	Median imagery complete	Median connectivity	Near misses / collisions
60	60 / 60	543 s	93.5 %	0 / 0

What the 60 simulations are

Five DJI Mavic-class UAVs operate from a forward ground control station (GCS) over a 2 km × 2 km disaster area. Every search site lies beyond the 600 m radio range of the GCS, so imagery must travel UAV-to-UAV over a multi-hop mesh. Each simulation is one saved scenario file (*scenarios/dev* seeds 0–29 and *scenarios/heldout* seeds 100–129) with:

- 7 random survey sites 0.5–2.4 km from the GCS and random initial batteries (45–100 %);
- a critical fault (hard: motor + radio, or failsafe: radio survives) on the busiest relay at a random time between 100 and 260 s (experiment **E1**);
- a new high-priority site reported mid-mission;
- a 15–60 s radio outage on another UAV;
- a 35–55 % energy-availability fault on the relay next to the GCS, forcing a relay handover.

All 60 use the default planner (**adaptive-spare**). The held-out 30 were never used to choose any parameter. The same files were also run with three other planners for comparison (section 5). A run **passes** only if it meets every predeclared internal constraint (section 3.8); completing the mission eventually is not enough.

Companion workbook: **results/UAV-X_60_simulations.xlsx**, with all 60 rows, live Excel formula checks against the constraint sheet, and the summary statistics that this report quotes. The pass/fail formulas in Excel agree with the simulator's own verdict in all 60 runs.

Scope and honesty

Simulation only. Flight is kinematic; the motor fault is an abstracted controlled descent; the energy fault is an instant loss of usable charge; the radio is a distance/loss abstraction with a single-channel airtime model; battery service is a 60 s swap (time-compressed). All thresholds are internal test constraints, not organiser rules.

1. System parameters (uavx/config.py)

Symbol	Meaning	Value
Δt	simulation step (= Webots basicTimeStep)	64 ms
N	UAVs in the fleet	5
v_h, v_z	cruise speed, climb/descent speed	15.0 m/s, 5.0 m/s
h_i	cruise layer of UAV i	$60 + 8(i - 1)$ m
R	radio range (link cut-off)	600 m
p_0, p_e	per-hop loss at 0 m, extra loss at the edge	0.01, 0.2
s_r	planned relay spacing	480 m (0.8 R)
c_m, c_h	battery drain moving / hovering	0.055 %/s, 0.05 %/s
E_{res}	landing reserve	15 %
C_{ch}	shared channel rate	4.0 Mbit/s
t_{mac}, t_{hop}	MAC overhead per transmission, forwarding latency per hop	0.5 ms, 20 ms
k_{max}	per-hop MAC retries	3
$B_{hb}, B_{cmd}, B_{thumb}, B_{chunk}$	packet sizes	200 B, 300 B, 8000 B, 32768 B
D	imagery per site	1.5 MB = 46 chunks
$T_{hb}, T_{cmd}, T_{lost}$	heartbeat and command period, GCS lost timeout	0.5 s, 0.5 s, 3.0 s
W	stability hold for 'recovered'	5.0 s
m_{ho}, m_{crit}	handover margin, critical margin	12.0 %, 7.0 %
d_{nm}, d_{col}	near-miss and collision-proxy distances	5.0 m, 1.5 m

2. Motion, energy and onboard autonomy

2.1 Kinematic flight

Each UAV first climbs to its own layer, then moves horizontally, then descends only over its pad or target, so transits never cross another layer.

$$\mathbf{x}_{t+\Delta t} = \mathbf{x}_t + \min(v_h \Delta t, \|\mathbf{g} - \mathbf{x}_t\|) \cdot (\mathbf{g} - \mathbf{x}_t) / \|\mathbf{g} - \mathbf{x}_t\| + \mathbf{a} \Delta t, z_{t+\Delta t} = z_t + \text{clip}(z^* - z_t - v_z \Delta t, v_z \Delta t)$$

$\mathbf{g}$ = horizontal goal, $\mathbf{a}$ = separation push (section 2.4), z^* = own layer h_i in transit, ground or pad height when descending; position hard-clamped to the geofence.

2.2 Battery and energy decisions

$$E_{t+\Delta t} = \max(0, E_t - c \Delta t), c = c_m \text{ if moving, } c_h \text{ if hovering}$$

$$E_{home}(\mathbf{x}) = (\|\mathbf{x}_{xy} - \mathbf{pad}\| / v_h + z / v_z) \cdot c_m + E_{res}$$

energy (in %) needed to fly home and land from $\mathbf{x}$, plus the 15 % reserve

$$\text{Onboard return-to-home: } E \leq E_{home}(\mathbf{x}) + 5 \%$$

taken onboard, overrides any GCS task

$$\text{Slot feasibility (GCS): } E > E_{transit} + T_{station} \cdot c_h + E_{home}(\text{slot})$$

$E_{transit} = (\text{climb}/v_z + \text{distance}/v_h) \cdot c_m$, $T_{station} = 120$ s

$$\text{Handover margin: } m = E - E_{home}(\mathbf{x}); \text{ start handover if } m < m_{ho}; \text{ leave regardless if } m < m_{crit}$$

$$\text{Battery swap: on landing, status = SERVICE for 60 s, then } E = 100 \%$$

2.3 Failsafe fault and lost-link rules

$$z_{t+\Delta t} = \max(z_{ground}, z_t - v_{fs} \Delta t), v_{fs} = 2.5 \text{ m/s}$$

critical fault: level controlled descent in place (abstracted); hard variant also switches the radio off

A UAV silent for more than 3 s while holding unacknowledged imagery becomes a *data mule* and flies toward the GCS **until every chunk is acknowledged** (hysteresis). An orphaned relay, and every UAV in its radio fragment, falls back toward the GCS after 3 s of silence.

2.4 Separation assurance

$$\text{vertical yield if } |\Delta x_h| < 60 \text{ m, } |\Delta z| < 12 \text{ m and } v_z \cdot (z_{other} - z) > 0; \text{ back off if } |\Delta z| < 8 \text{ m and } |\Delta x_h| < 25 \text{ m}$$

$\mathbf{a} = 6 \cdot (d_{\min} - d)/d_{\min} \cdot \mathbf{u}$ for $d < d_{\min} = 5$ m; keep-out radius 15 m around a failsafe descent column

$\mathbf{u}$ = unit vector away from the other UAV (a fixed per-UAV direction if exactly stacked)

3. Radio, routing and the shared channel

3.1 Link model

$p(d) = p_0 + p_e (d/R)^4$ for $d \leq R$, $p(d) = 1$ for $d > R$ ($p_0 = 0.01$, $p_e = 0.2$, $R = 600$ m)

d = 3D distance between the two radios; p = probability one transmission attempt is lost

degraded profile: $R = 480$ m, $p_e = 0.35$; *shadowed*: $p \leftarrow \min(1, p + 0.3(1 - c/25))$ if terrain clearance $c < 25$ m, *blocked* if $c < 0$

uncalibrated stress cases used only for sensitivity runs

3.2 Minimum-ETX routing

$ETX(a,b) = 1 / (1 - p(d_{ab}))$, $cost(n) = \min_{paths} \sum_{hops} ETX$

Dijkstra from the GCS every step; a relay being drained (handover) may not forward

$P_{path} = \prod_{hops} (1 - p_i)$

probability one packet survives every hop without retries

3.3 Airtime, latency and retries

$T_{air} = 8B / C_{ch} + t_{mac}$ e.g. a 32 KiB chunk: 66.0 ms per hop

the sender and every radio neighbour are busy for T_{air} (carrier sense); distant nodes can transmit in parallel

$t_{arrive} = t_{start} + T_{air} + t_{hop}$, $t_{start} = \max(t_{ready}, \text{medium free})$

hop success after at most $k_{max} + 1 = 4$ attempts: $P = 1 - p^4$

each attempt costs airtime and is counted as a transmission

loss draw $u = \text{splitmix64}(\text{seed}, \text{link}, [t/\Delta t], \text{attempt}) / 2^{53}$; lost if $u < p$

common random numbers: every planner sees the same channel realisation on the same link and slot

3.4 Imagery transfer

$N_{chunks} = \lceil D / B_{chunk} \rceil = \lceil 1.5 \cdot 10^6 / 32768 \rceil = 46$

sliding window of 4 chunks, end-to-end acknowledgement, retransmit after 4 s; a site is complete when all chunks are at the GCS

3.5 Delivery ratio (PDR)

$PDR = |\{ \text{packets delivered within their deadline} \}| / |\{ \text{packets generated} \}|$

per class (heartbeat, command, ack, thumbnail); only packets whose 2 s deadline has passed; a packet generated with no route counts as lost; empty set $\rightarrow$ N/A, never 100 %

$generated = delivered + expired + dropped + dropped_{radio} + lost + pending$

exact accounting identity, checked by the test suite

3.6 GCS planner

$n_{relays} = \max(0, \lceil d_{anchor \rightarrow site} / s_r \rceil - 1)$, relay k at $\mathbf{a} + k/(n+1) \cdot (\mathbf{site} - \mathbf{a})$

chains branch from the nearest existing anchor (GCS or planned relay), so relays form a tree

$assignment = \text{argmin}_{perm} \sum \| \mathbf{x}_{uav} - \mathbf{slot} \| - 30 \text{ m} \cdot [\text{same slot as before}]$

exhaustive over $\leq 5! = 120$ permutations, subject to the energy feasibility of section 2.2

silent surveyor kept for $\min(240 \text{ s}, 2 \cdot d_{last \rightarrow site} / v_h + 10 \text{ s} + 60 \text{ s})$

only if its heartbeat had acknowledged that survey task; otherwise the site is re-queued at once

3.7 Recovery after a fault (evaluator)

Populations are frozen at the fault time t_0 : **orphaned** = UAVs that had a route before the fault and lost it; **required** = orphaned (+ the faulted UAV itself if the fault is a temporary radio outage). An aircraft loss is irrecoverable and never counted as recovered.

$$t_{first} = \min\{ t > t_0 : \exists u \in required \text{ with a route } \} \quad t_{all} = \min\{ t : \forall u \in required \text{ have a route } \}$$

$$t_{stable} = \min\{ t : \forall \tau \in [t, t + W], \forall u \in required \text{ have a route } \}, \quad t_{confirmed} = t_{stable} + W, \quad W = 5 \text{ s}$$

any disconnection resets the hold; all times reported relative to t_0 ; 'none orphaned' → N/A, not 0

$$disconnected_{fault} = \sum_{steps} |\{ u \in required \text{ without a route } \}| \cdot \Delta t \text{ (UAV·s)}$$

3.8 Network and mission metrics

$$Connectivity = \text{mean}_t [|airborne \text{ UAVs with a route}| / |airborne \text{ UAVs}|] \times 100 \%$$

$$Disconnected \text{ UAV·s} = \sum_t |airborne \text{ UAVs without a route}| \cdot \Delta t$$

$$state(t) \in \{ partitioned, connected \text{ with a critical relay}, connected \& redundant \}$$

critical relay = a UAV whose removal alone disconnects another routed UAV (articulation point); shares are of time with ≥ 1 airborne UAV

$$survey \rightarrow GCS \text{ latency} = t_{all \text{ chunks at GCS}} - t_{survey \text{ done}}; \quad high\text{-priority response} = t_{imagery \text{ at GCS}} - t_{report}$$

$$operational \text{ success} = [T_{img} \leq 900 \text{ s}] \wedge [T_{land} \leq 1200 \text{ s}] \wedge [T_{HP} \leq 240 \text{ s}] \wedge [f_{60s} \geq 0.75] \wedge [d_{min} \geq 5 \text{ m}] \wedge [no \text{ collision}] \wedge [no \text{ geofence breach}] \wedge [no \text{ depletion}] \wedge [E_{landing} \geq 15 \%]$$

predeclared internal constraints (config.CONSTRAINTS); implemented identically in the Excel check columns

3.9 Safety: continuous closest approach

Between two samples both UAVs move linearly. With relative position $\mathbf{r}_0$ and relative displacement $\Delta \mathbf{r}$ over the step:

$$s^* = clip(-(\mathbf{r}_0 \cdot \Delta \mathbf{r}) / \|\Delta \mathbf{r}\|^2, 0, 1), \quad d_{min} = \|\mathbf{r}_0 + s^* \Delta \mathbf{r}\|$$

catches crossings that happen between 64 ms samples; consecutive steps below 5 m form one near-miss episode

3.10 Onboard person counting (perception)

For a person box with bottom-centre pixel (u, v) in a $W \times H$ image, camera yaw ψ , downward pitch θ and horizontal field of view ϕ :

$$f_x = (W/2) / \tan(\phi/2), \quad a = -(u - W/2)/f_x, \quad b = -(v - H/2)/f_x$$

$$\mathbf{ray} = \mathbf{f} + a \mathbf{l} + b \mathbf{u}, \quad \mathbf{f} = (\cos\theta \cos\psi, \cos\theta \sin\psi, -\sin\theta), \quad \mathbf{l} = (-\sin\psi, \cos\psi, 0), \quad \mathbf{u} = (\sin\theta \cos\psi, \sin\theta \sin\psi, \cos\theta)$$

$$\mathbf{p}_{ground} = \mathbf{c} + \lambda \mathbf{ray}, \quad \lambda = (z_{ground} - c_z) / ray_z$$

c = camera position; measured projection error: median 0.06 m, worst 0.12 m

$$count = |\{ tracks \text{ within 8 m of the site, seen in } \geq 20 \% \text{ of settled frames } \}|, \quad a \text{ detection joins a track if within 0.3 m}$$

benchmark (72 views, matched to true positions): precision = $TP/(TP+FP) = 0.992$, recall = $TP/(TP+FN) = 0.903$

3.11 Worked examples (numbers from the actual code)

Quantity	Formula with numbers	Result
Loss on one 480 m relay hop	$0.01 + 0.2 \cdot (480/600)^4$	0.092 (9.2 %)
ETX of that hop	$1 / (1 - 0.092)$	1.101 transmissions
Loss at the range edge (600 m)	$0.01 + 0.2 \cdot 1^4$	0.210
Hop failure after all MAC retries (480 m)	0.092^4	7.14e-05
Relays to reach a site 2.4 km away	$\lceil 2400/480 \rceil - 1$	4 relays (5 hops)
Heartbeat airtime per hop	$8 \cdot 200/4 \cdot 10^6 + 0.5 \text{ ms}$	0.90 ms
Imagery chunk airtime per hop	$8 \cdot 32768/4 \cdot 10^6 + 0.5 \text{ ms}$	66.0 ms
One chunk over the 5-hop chain	$5 \cdot (66.0 + 20) \text{ ms}$	0.43 s
Whole site (46 chunks), shared channel, 5 hops	$46 \cdot 5 \cdot 66.0 \text{ ms} / 4 \text{ in parallel (lower bound)}$	$\approx 3.8 \text{ s}$
Energy to return from (900, 900) at 60 m	$(2459/15 + 60/5) \cdot 0.055 + 15$	24.7 %
Endurance at cruise from 100 % to the reserve	$(100 - 15) / 0.055$	25.8 min
Stability confirmation	$t_{\text{stable}} + 5 \text{ s}$	e.g. 15.1 → 20.2 s (demo run)

3.12 Where each formula lives (controllers/swarm_supervisor/)

Model	Source	Function
Kinematic flight, layers, yielding	uavx/agent.py	Uav.fly_to, _choose_target
Battery, return-home, swap	uavx/agent.py	energy_to_home_from, step, _on_pad
Slot energy feasibility	uavx/agent.py	energy_for_slot
Link loss, ETX routing, critical relays	uavx/comms.py	loss_prob, Network.update, critical_relays
Airtime, queues, retries, PDR	uavx/comms.py	Channel.step, cohort_pdr
Relay chains, assignment, handover	uavx/gcs.py	_plan, _assign, _handover, _retire
Silent-surveyor rule	uavx/gcs.py	_out_of_contact, _expected_silence
Separation assurance	uavx/sim.py	_collision_avoidance
Recovery, connectivity, safety metrics	uavx/metrics.py	FaultTracker, Metrics, seg_min_dist
Constraints	uavx/sim.py	evaluate_constraints
Ground projection, person tracks	perception.py	project, SiteCounter

4. Results of all 60 simulations

E1 (critical fault on the busiest relay), planner adaptive-spare. Columns: Img = all imagery at GCS; Land = surviving fleet landed; HP = high-priority report→imagery; 60s = share of sites whose imagery arrived ≤ 60 s after survey; Conn = connectivity; Part / Crit = % time partitioned / with a critical relay; Disc = disconnected UAV-s; PDR = heartbeat PDR; Orph = UAVs orphaned by the fault; Stab = stable recovery confirmed (- = none orphaned); Sep = minimum separation; Bat = lowest landing battery; HO = energy handovers; OK = passed every constraint.

4.1 Development scenarios (1-30)

#	Scenario	Fault	Img s	Land s	HP s	60s %	Conn %	Part %	Crit %	Disc UAV-s	PDR %	Orph	Stab s	Sep m	Bat %	HO	OK
1	seed 000	failsafe	453	581	43	88	92.6	16.8	54.3	158	94.1	0	-	6.0	21	0	PASS
2	seed 001	hard	676	780	68	100	92.3	21.2	60.6	219	93.1	4	27.2	6.0	25	1	PASS
3	seed 002	failsafe	351	482	44	88	94.3	13.5	50.6	107	95.4	0	-	6.0	33	0	PASS
4	seed 003	hard	533	634	108	100	94.4	19.4	50.7	112	95.0	0	-	6.0	26	1	PASS
5	seed 004	hard	502	654	48	100	96.0	11.4	66.2	96	95.8	4	14.5	6.0	34	0	PASS
6	seed 005	failsafe	566	661	92	88	92.0	28.8	50.5	173	93.7	0	-	6.0	26	1	PASS
7	seed 006	hard	476	605	93	100	94.3	20.4	56.9	148	93.3	3	13.1	6.0	33	0	PASS
8	seed 007	hard	514	660	24	88	92.1	18.4	59.9	204	92.0	2	27.6	6.0	32	0	PASS
9	seed 008	hard	638	739	45	100	90.0	35.8	37.7	260	91.0	0	-	6.0	25	1	PASS
10	seed 009	hard	547	678	102	100	94.0	13.4	65.6	153	94.1	4	28.5	6.0	31	0	PASS
11	seed 010	failsafe	634	730	102	100	91.3	26.0	59.1	248	92.3	0	-	8.0	22	1	PASS
12	seed 011	failsafe	300	371	72	100	99.0	4.0	46.6	12	99.4	0	-	6.0	41	0	PASS
13	seed 012	hard	454	564	55	88	91.2	28.2	40.4	204	91.1	4	20.2	6.0	26	0	PASS
14	seed 013	failsafe	463	614	50	100	99.1	3.6	68.7	20	99.2	0	-	6.0	34	0	PASS
15	seed 014	failsafe	410	558	46	100	96.7	13.0	57.8	73	97.3	0	-	6.0	44	0	PASS
16	seed 015	hard	529	664	37	100	97.6	6.7	71.3	58	97.4	4	10.8	6.0	33	0	PASS
17	seed 016	failsafe	360	445	73	100	95.1	13.4	46.5	75	96.3	0	-	6.0	35	0	PASS
18	seed 017	failsafe	586	722	124	100	98.0	8.6	70.9	62	98.1	0	-	6.0	34	0	PASS
19	seed 018	hard	622	730	118	100	90.5	27.3	51.5	270	91.2	4	22.1	6.0	25	1	PASS
20	seed 019	failsafe	407	525	65	100	95.6	17.1	48.7	82	96.6	0	-	6.0	42	0	PASS
21	seed 020	hard	399	494	81	100	92.4	30.2	21.0	131	92.5	0	-	6.0	24	0	PASS
22	seed 021	failsafe	411	530	54	100	98.2	7.4	60.0	35	98.5	0	-	6.0	20	0	PASS
23	seed 022	hard	707	848	81	100	97.7	9.7	76.9	82	97.1	0	-	6.0	27	0	PASS
24	seed 023	hard	746	848	116	100	93.7	23.0	57.1	181	94.3	0	-	6.0	25	1	PASS
25	seed 024	failsafe	539	683	37	100	92.8	21.2	62.1	201	93.4	0	-	6.0	31	0	PASS
26	seed 025	failsafe	386	483	80	100	91.6	30.3	35.9	141	93.5	0	-	6.0	35	0	PASS
27	seed 026	hard	468	566	85	100	90.9	22.6	49.6	190	91.9	4	30.3	6.0	33	0	PASS
28	seed 027	failsafe	760	862	105	100	94.5	22.1	56.1	177	95.4	0	-	6.0	25	1	PASS
29	seed 028	hard	678	812	146	100	97.4	10.3	72.5	78	97.1	0	-	6.0	27	0	PASS
30	seed 029	hard	532	667	128	100	90.7	16.5	64.3	248	90.7	4	37.1	6.0	31	0	PASS

4.2 Held-out scenarios (31-60)

#	Scenario	Fault	Imgs	Land s	HP s	60s %	Conn %	Part %	Crit %	Disc UAV-s	PDR %	Orph	Stab s	Sep m	Bat %	HO	OK
31	seed 100	failsafe	533	672	23	100	97.7	9.3	68.8	57	98.1	0	-	6.0	28	0	PASS
32	seed 101	failsafe	700	844	109	100	92.4	20.8	64.8	240	94.1	0	-	6.0	22	1	PASS
33	seed 102	failsafe	509	652	179	100	93.1	16.6	60.9	164	94.7	0	-	6.0	28	0	PASS
34	seed 103	failsafe	567	669	98	100	97.0	11.5	62.9	76	97.5	0	-	6.3	22	1	PASS
35	seed 104	failsafe	468	606	89	100	97.0	11.9	59.4	65	97.5	0	-	6.0	37	0	PASS
36	seed 105	failsafe	624	751	112	100	93.3	21.5	55.8	185	94.3	0	-	6.0	24	1	PASS
37	seed 106	hard	776	885	218	88	93.1	22.3	62.1	230	93.5	3	20.7	6.0	24	1	PASS
38	seed 107	failsafe	738	880	53	100	91.7	25.8	58.9	257	93.7	0	-	6.0	22	1	PASS
39	seed 108	failsafe	639	767	102	100	96.1	13.9	67.9	109	96.9	0	-	6.0	30	0	PASS
40	seed 109	failsafe	696	831	32	100	92.2	29.4	53.5	244	93.5	0	-	6.0	26	1	PASS
41	seed 110	hard	466	602	80	100	91.4	24.2	51.1	186	92.4	4	20.8	6.0	35	0	PASS
42	seed 111	hard	502	626	59	100	95.1	13.0	57.1	122	94.6	4	15.5	6.0	35	0	PASS
43	seed 112	failsafe	498	632	62	100	91.3	32.3	38.8	238	91.6	0	-	6.0	27	1	PASS
44	seed 113	failsafe	742	849	74	100	96.4	14.6	66.6	115	96.8	0	-	6.0	24	1	PASS
45	seed 114	failsafe	704	824	93	88	74.9	45.4	36.4	708	79.9	0	-	6.0	22	1	PASS
46	seed 115	failsafe	645	754	107	100	92.6	22.3	53.2	193	94.5	0	-	6.0	25	1	PASS
47	seed 116	failsafe	872	1007	98	100	92.9	24.3	60.3	262	94.0	0	-	6.0	26	1	PASS
48	seed 117	hard	564	702	128	100	97.0	5.6	72.2	81	96.5	4	20.2	6.0	30	0	PASS
49	seed 118	failsafe	448	559	55	100	95.5	19.5	53.2	98	96.3	0	-	6.0	29	0	PASS
50	seed 119	hard	682	779	115	88	93.8	25.2	55.0	196	93.3	0	-	6.0	24	1	PASS
51	seed 120	hard	629	760	113	100	91.8	20.2	61.2	220	92.9	0	-	6.0	25	1	PASS
52	seed 121	failsafe	485	592	142	100	95.0	19.8	51.9	106	95.9	0	-	6.0	23	1	PASS
53	seed 122	failsafe	686	778	74	88	92.7	24.1	54.9	188	94.4	0	-	6.0	23	1	PASS
54	seed 123	hard	821	920	200	100	89.5	32.9	52.6	340	91.1	4	24.1	6.0	25	1	PASS
55	seed 124	failsafe	550	700	29	100	93.3	22.6	56.3	144	95.3	0	-	6.0	27	1	PASS
56	seed 125	failsafe	590	717	141	100	97.8	9.0	68.3	59	98.1	0	-	6.0	26	1	PASS
57	seed 126	hard	478	599	126	100	94.0	14.4	56.2	129	94.3	4	28.1	6.0	31	0	PASS
58	seed 127	failsafe	378	524	55	88	81.9	44.2	27.6	334	85.2	0	-	6.0	27	1	PASS
59	seed 128	failsafe	355	420	45	88	88.8	37.3	18.3	189	90.4	0	-	6.0	23	1	PASS
60	seed 129	hard	600	736	118	100	96.6	7.3	74.1	99	96.1	4	21.6	6.0	64	0	PASS

All 60 runs: 0 near misses (< 5 m), 0 collision proxies (< 1.5 m), 0 geofence violations, 0 battery depletions and 0 information-flow bypasses. Faults marked *hard* also switch the faulted UAV's radio off, so the GCS can only detect them by the 3 s heartbeat timeout.

4.3 Summary statistics (computed by formulas in the workbook)

Metric	Mean	Median	Best	Worst	Dev mean	Held-out mean	n
Imagery complete (s)	559.9	542.8	300.2	871.5	521.6	598.1	60
Fleet landed (s)	680.5	670.4	371.1	1007.0	639.7	721.2	60
High-priority response (s)	87.5	83.3	23.3	217.7	77.3	97.6	60
Sites within 60 s (fraction)	0.977	1.000	1.000	0.875	0.979	0.975	60
Connectivity (%)	93.5	93.5	99.1	74.9	94.2	92.9	60
Time partitioned (%)	19.7	20.0	3.6	45.4	18.0	21.4	60
Time with a critical relay (%)	55.8	57.0	18.3	76.9	55.7	56.0	60
Time redundant (%)	24.5	22.3	49.4	13.4	26.3	22.6	60
Disconnected UAV-s	163.8	155.2	12.4	707.6	139.8	187.7	60
Heartbeat PDR (%)	94.3	94.3	99.4	79.9	94.7	93.9	60
Command ack p95 (s)	0.616	0.616	0.580	0.703	0.620	0.612	60
UAVs orphaned by fault	1.1	0.0	0.0	4.0	1.2	0.9	60
Stable recovery confirmed (s)	22.5	21.6	10.8	37.1	23.1	21.6	17
Fault UAV-s disconnected	18.3	0.0	0.0	127.7	22.3	14.3	60
Min separation (m)	6.0	6.0	8.0	6.0	6.1	6.0	60
Min landing battery (%)	28.9	26.7	63.6	20.1	30.1	27.7	60

Constraint check (Excel formula)	All 60	Dev (of 30)	Held-out (of 30)
Imagery complete within limit	60	30	30
High-priority response within limit	60	30	30
Timely sites >= minimum	60	30	30
Separation >= 5 m, no near miss	60	30	30
Landing battery >= reserve	60	30	30
No collision / depletion; fleet landed in time	60	30	30
ALL CHECKS (operational success)	60	30	30

Total near misses (< 5 m): **0** · Total collisions (< 1.5 m proxy): **0** · Total battery depletions: **0** · Total energy handovers: **28**

Best/worst: for lower-is-better metrics best = minimum and worst = maximum, and the reverse for higher-is-better ones. The stable-recovery row covers only the runs in which the fault orphaned at least one UAV (n shown); in the other runs nothing had to recover.

5. Same 60 scenarios, four planners

adaptive-spare (default): relay chains re-planned every second, a backup relay only when a UAV is spare.
 adaptive-reserve: always holds one UAV as a backup relay. adaptive-conditional: holds a backup only for a relay carrying ≥ 3 UAVs. fixed-baseline: relay backbone assigned once and never re-planned. Identical UAVs, onboard rules, channel, energy and safety logic. Cells: median (worst).

Development (30)	adaptive-spare	adaptive-reserve	adaptive-conditional	fixed-baseline
Passed every constraint	30/30	30/30	28/30	4/30
Eventually completed	30/30	30/30	30/30	22/30
All imagery at GCS (s)	514.1 (760.3)	619.9 (824.6)	560.3 (1066.0)	738.7 (2263.3)
High-priority response (s)	73.2 (145.6)	81.3 (128.1)	79.4 (142.9)	361.7 (1899.8)
Connectivity (%)	94.0 (90.0)	91.8 (85.9)	93.4 (79.7)	43.4 (19.9)
Disconnected UAV-s	141.4 (269.6)	187.0 (372.7)	161.3 (343.2)	1702.3 (7192.7)
UAVs orphaned by fault (mean)	1.2	0	0.5	1.1
Stable recovery confirmed (s)	22.1 (37.1)	none orphaned	19.0 (23.2)	52.1 (70.0)
Near misses · collisions	0 · 0	0 · 0	0 · 0	0 · 0

Held-out (30)	adaptive-spare	adaptive-reserve	adaptive-conditional	fixed-baseline
Passed every constraint	30/30	27/30	30/30	4/30
Eventually completed	30/30	30/30	30/30	27/30
All imagery at GCS (s)	589.9 (871.5)	640.7 (907.9)	645.6 (891.0)	828.0 (2356.3)
High-priority response (s)	98.1 (217.7)	90.8 (287.7)	95.0 (227.8)	294.1 (757.7)
Connectivity (%)	93.1 (74.9)	91.8 (85.6)	92.6 (76.2)	44.6 (17.2)
Disconnected UAV-s	185.2 (707.6)	203.3 (440.8)	160.5 (581.1)	1813.8 (7591.9)
UAVs orphaned by fault (mean)	0.9	0	0.2	0.7
Stable recovery confirmed (s)	20.8 (28.1)	none orphaned	6.5 (8.6)	66.4 (103.3)
Near misses · collisions	0 · 0	0 · 0	0 · 0	0 · 0

Before (recorded v3 code) → after (v3.1), same 60 scenarios, adaptive-spare

Metric	Before	After
All imagery at GCS, median (s)	562.3	542.8
Connectivity, median (%)	91.3	93.5
Disconnected UAV-s, median	235.0	155.2
UAVs orphaned by fault, mean	1.4	1.1
Minimum separation, worst (m)	6.0	6.0

6. Robustness beyond the 60 runs

Radio stress profiles (development scenarios, adaptive-spare):

Radio profile	Passed	Connectivity median (%)	Imagery median (s)	Imagery worst (s)
simple (the 60 runs, dev half)	30/30	94.0	514	760
shadowed	30/30	93.8	537	775
degraded	13/30	70.8	880	1556

Fault on the same aircraft id in every planner (E2): adaptive-spare passes 29/30 (dev) and 29/30 (held-out). The failures are high-priority responses of 272–276 s when a hard fault hits the UAV assigned to the high-priority site while it is still in radio range: the GCS cannot tell this from a normal survey leg and waits for the expected out-and-back time.

7. Limitations

- Kinematic flight: no rotor or aerodynamic dynamics; the motor fault is an abstracted controlled descent.
- Energy fault = instant loss of usable charge; no cell voltage/current model. Battery swap is time-compressed (60 s).
- Radio: distance/loss model with a single-channel carrier-sense airtime abstraction; no hidden-terminal or interference model; the stress profiles are uncalibrated.
- Perception: stock COCO YOLOv8n; people standing closer than about 0.5 m are merged into one box (recall 0.90 in the benchmark).
- All thresholds are internal. This is a simulation study, not hardware validation or a competition-readiness claim.

8. Reproduce

```
python tools/evaluate.py # the 60 runs (and the other planners)
python tools/make_report_xlsx.py # the workbook
python tools/make_report_pdf.py # this report
python -m unittest discover -s tests # 65 tests
```

Sources: results/evaluation/E1-busiest__dev.json, E1-busiest__heldout.json, E2-fixed__*.json, E3-radio-*__dev.json, results/before_after.json, results/UAV-X_60_simulations.xlsx; model: controllers/swarm_supervisor/uavx/*.py and perception.py.